

FINAL DESIGN PROJECT

The final due date for the project is May 13th, 2026, at 5pm. No extensions will be granted. There will be design reviews on the weeks of April 20th and May 4th. Every group must be present in the 1st design review. 2nd design review will be arranged as project dedicated reservable office hour slots.

Task

The goal of the project is to design a fully differential switched capacitor amplifier including the OTA, MOS switches and capacitors.

Groups

You should form groups of 3 students who will collaborate on the project. Groups of four students will not be allowed; groups of two students require instructor's approval. Discussion with others is encouraged but genuine design work from each group is required. All sources used need to be properly referenced or acknowledged in the project report. Make sure to interact regularly with the TA during office hours/design reviews about your progress, ideas and challenges.

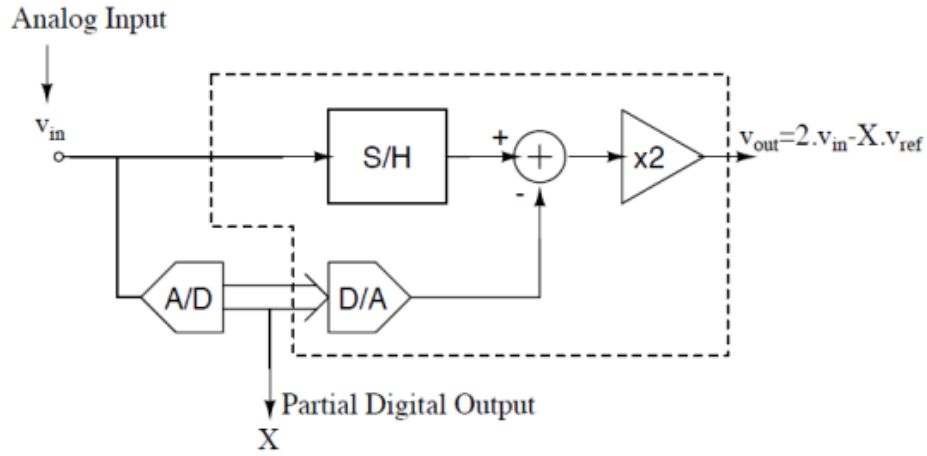
Background

The project assignment is inspired by the design of amplifiers for pipelined analog-to-digital converters (ADCs). It is not strictly necessary to understand the background material in detail but see the accompanying presentation for an example on how pipelined ADCs work.

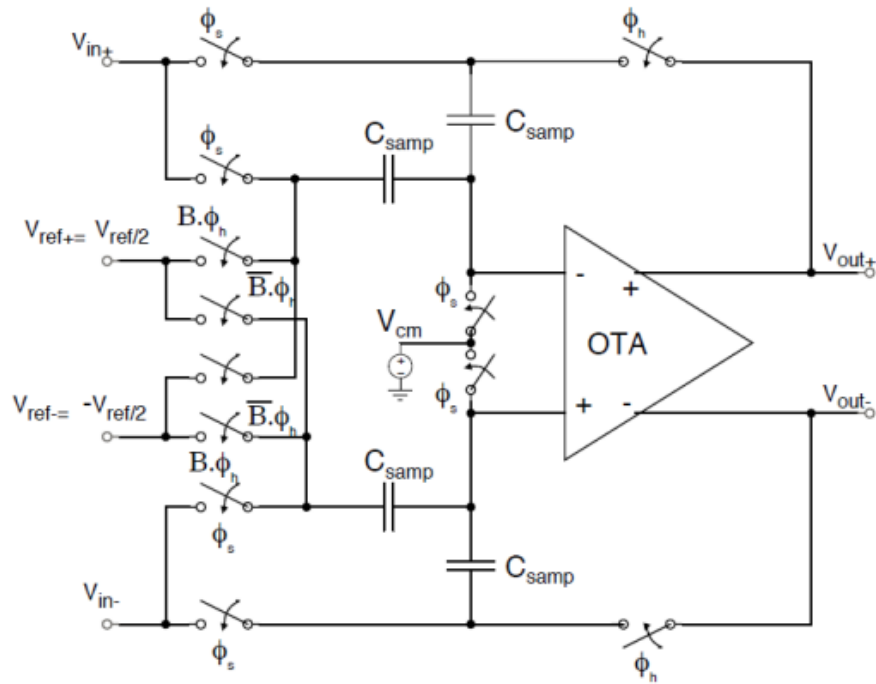
Each stage in the pipeline does a coarse A/D conversion to determine the partial digital output X and computes the residue voltage, v_{out} , to be processed by later stages. A basic block diagram of a stage in a 1.5bit/stage pipelined A/D converter is shown in Fig. 1(a). The A/D determines if $v_{in} < -v_{REF}/4$, if $-v_{REF}/4 < v_{in} < v_{REF}/4$, or if $v_{in} > v_{REF}/4$, and outputs $X = -1$, $X = 0$ and $X = 1$ respectively. The residue v_{out} is computed as $2(v_{in} - X \cdot v_{REF}/2)$ in the analog domain.

In modern implementations, the entire circuit (and hence the OTA) is fully differential. An example of a possible implementation of the dashed area in Fig. 1(a) is shown Fig. 1(b). There are many different implementations of this block - each having its own advantages with respect to sampling, charge injection, parasitics, offset etc.

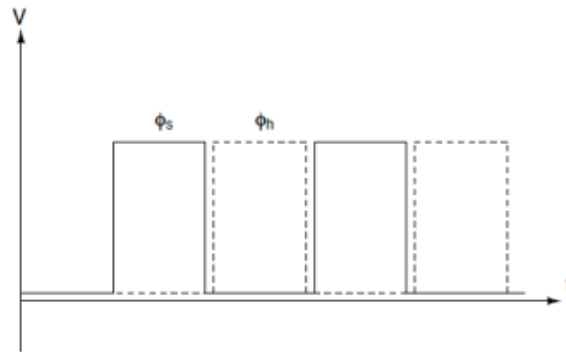
For your own understanding, you might consider first analyzing the block diagram (Fig. 1(a)) and see for yourself how a pipelined A/D works. Due to the amplification in each stage, the accuracy of the complete A/D converter is strongly dependent on the accuracy of the first and second stage, in particular on the accuracy of the dashed area of the first stage of the pipelined converter. Now, for your own understanding, analyze the circuit (Fig. 1(b)) shown and see that indeed it does fit into the dashed area of the block diagram.



(a)



(b)



(c)

Figure 1: Background (a) Block diagram of a stage in a pipelined ADC (b) Possible architecture for a stage in 1.5bit/ stage pipelined ADC (c) Sample and Hold Clock Phases

Partial Digital Output & Control Bits		
X	B	B
-1	0	1
0	0	0
1	1	0

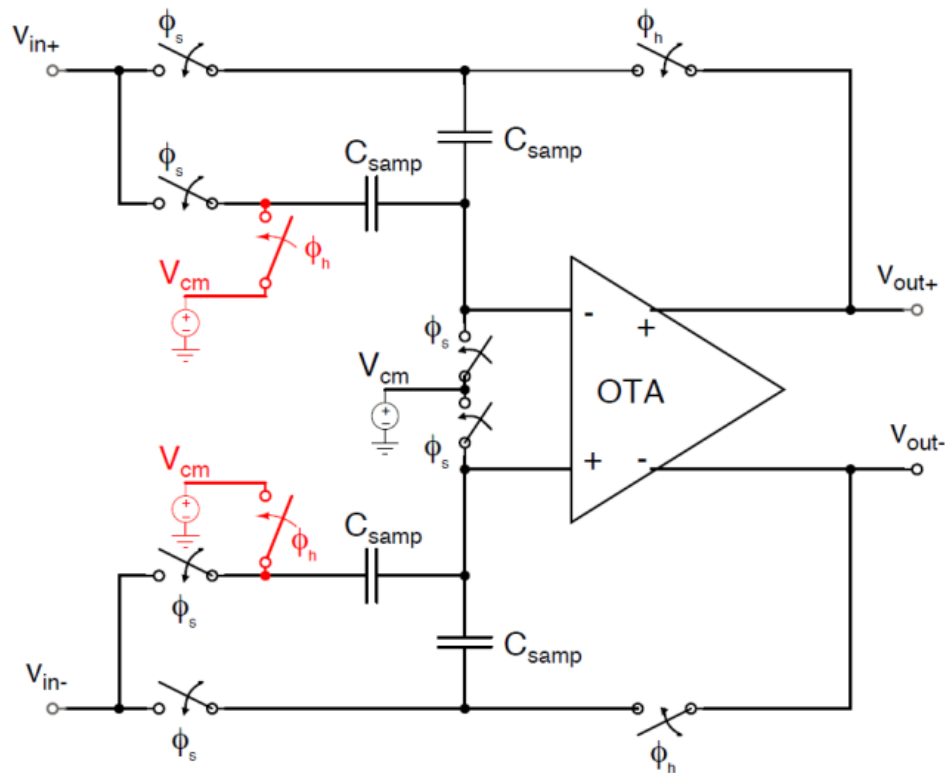


Figure 2: Circuit to be designed in this project: Fully differential switched capacitor sample-and-hold amplifier with a nominal gain of 2

References

For further reading on pipelined A/D Converters, or to get a clear understanding, please refer to the following papers:

1. "A pipelined 5-Msample/s 9-bit analog-to-digital converter," S. H. Lewis & Paul Gray: IEEE Journal of Solid-State Circuits Dec. 1987
2. "A 12-bit 1-Msample/s capacitor error-averaging pipelined A/D converter," Song, Tompsett & Lakshmikumar: IEEE Journal of Solid-State Circuits Dec. 1988
3. You can also refer to Johns and Martin, Analog Integrated Circuit Design (Second Edition), Chapter 17, Section 17.4 and 17.6 for a very brief introduction on pipelined ADCs.
4. For fundamentals on A/D conversion, definitions etc, refer to Johns and Martin, Analog Integrated Circuit Design (Second Edition), Chapter 15

Project Design Goal

Design the fully differential switched-capacitor sample-and-hold amplifier with a nominal gain of 2 shown on Fig. 2. Minimize the power consumption while meeting all the design specifications for typical process and operating conditions.

Process and Typical Operating Conditions

- Nominal VDD: 1.8 V.
- Nominal Temperature: 300 K.
- Nominal Process: TSMC018 Teaching model - tt corner.

System Parameters Selection

Each group will complete a design to function in a pipelined ADC with a specific set of speed and accuracy specifications. A set of sampling frequency – total ADC resolution pairs is given to class to choose from. Use [sign-up sheet](#) to enter your group information and select parameter set.

Sampling frequency (f_s): 4MHz – 8MHz – 16MHz

Total ADC resolution: 7bits – 8bits – 9bits

Deadlines and Deliverables

The project is split into 3 steps:

1. System-Level Design:

Design Review: week of April 20th, 2026

2. OTA Transistor-Level Design:

Design Review: week of May 4th, 2026

3. Complete Design and Final Report:

Submission Deadline: May 13th, 2026 (5 pm. No extensions.)

There will be design review sessions for the first two stages of the project. Each group must attend the 1st design review to discuss their design with the TA for their project. Characterize and document your circuit details and progress for each design review for efficient meeting. Suggested to include following in the documentation:

- Transistor level schematics of the design. Tabulate all component values and transistor sizes.
- Top-level schematic for all test setups. Create a symbol for your OTA and then create top-level test benches.
- Simulation plots for typical operation.

The grading is based on every stage of the project. Mandatory design review carries a grade of 20 points. The final report carries 80 points. Project report files for the final submission must be zipped and emailed to the TA. Please follow the instructions (below) for the final report exactly.

Highlighted progress by each design review is the minimum. Groups can progress further and discuss more details in review.

Final Report

- For your final report, please arrange all simulation plots and tabular results in one PDF file, briefly explain/comment when appropriate. Cadence plots and schematics usually do not look clear on reports. You should use Xcircuit for making schematics and MATLAB for plotting. Do not print pages of simulations and schematics but rather select the key simulations to show and extract key performance metrics and organize them.
- Performance summary of the individual blocks and overall design in a datasheet. Tabulate performance parameters (DC gain, phase margin and bandwidth of the fully differential OTA and FOM, power consumption of the overall design etc.).
- For the OTA design, you need to submit the following simulation results:
 1. Differential frequency response of OTA indicating DC gain, bandwidth and phase margin.
 2. Common mode frequency response of OTA.
 3. Transient differential and common mode step response for an input step of amplitude V_{LSB} (size of the least significant bit) and for an input step of amplitude V_{FS} (full-scale voltage).

• For the overall design including the OTA, capacitors and switches, you need to submit the following simulation results:

1. Transient simulations to prove the 2x amplification happens with the right accuracy; show that you have enough settling time, etc.
2. Transient simulations for a small signal step at input, going from V_{LSB} to $-V_{\text{LSB}}$ and back to V_{LSB} .
3. Transient simulations for a ramp with a period of $1/20f_s$, covering the full-scale range of the design $[-V_{\text{FS}}, V_{\text{FS}}]$.
4. Transient simulations for a large signal step at input, going from V_{FS} to $-V_{\text{FS}}$ and back to V_{FS} .

You need to hand in a clear report. It should allow the educated reader to understand how the design was performed. The report needs to focus on the design procedure that you followed. Make sure to include the trade-offs considered, steps taken to optimize power consumption, reason for the choice of circuit topology etc. You are not graded based only on the performance of your design, but also on explaining what design techniques you used to achieve the reported results or what design challenges limited you from further improving the performance.

The length of the report **must not exceed 6 pages** (not including figures/ tables). It should contain all circuit schematics, mention sizes of all components and include important simulation plots. The figures and the tables should be at the end of the report, and large enough so everything is readable.

Figure of Merit

For the final report, you need to normalize your power consumption for operating speed and accuracy by computing the figure of merit (FOM) of your design given by:

$$\text{FOM} = \text{Power} / (2^{\text{ENOB}} f_s)$$

where f_s is the sampling speed, the effective number of bits, ENOB, is defined as $(\text{SINAD}-1.76)/6.02$ where SINAD is the maximum signal to noise-and-distortion ratio.

By normalizing the performance with a figure of merit, the performance of different designs can be compared. In particular, the figure of merit assumes that doubling the sampling speed should result in doubling of power.

Similarly, increasing the number of bits by 1, i.e. improving the SINAD by 6 dB, should require twice the amount of power. This is a generally used figure of merit in literature, however, there are limitations as to how accurate or applicable some of the design normalization assumptions are.

For ease of calculation, you can plug in the number of bits specified for your team in the FOM equation as ENOB.

Design Procedure

The following subsections give an idea of where to start. The final goal is to meet the overall requirement of the system and not for individual blocks. Expect that a few iterations may be required before you meet the specifications.

1. System-Level Design (Week of April 20th, 2026)

- Choose a reasonable value for full-scale voltage (V_{FS}) and common mode voltage (V_{CM}) for a supply voltage of V_{DD} shown in Fig. 3 and calculate the size of least significant bit (V_{LSB}). What constitutes a reasonable V_{FS} or V_{CM} ?

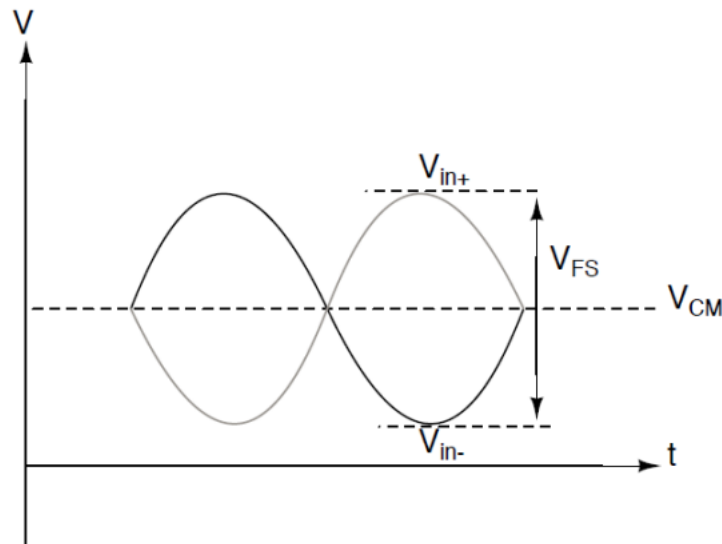


Figure 3: Full Scale Voltage and Common Mode Voltage Definition

A. Switched-Capacitor Input Stage

- In a sampled system, thermal noise due to the switch resistor is sampled on the capacitor every cycle. This sampled noise can be derived to be $V_{RMS}^2 = kT/C$, where k is the Boltzmann constant. Choose the sampling capacitor (C_{smp}) size such as the sampling noise $V_{RMS}^2 < V_{LSB}^2/24$ for your design.
- Derive the transfer function for the switched capacitor gain block.
- Assume an ideal OTA (very large Gain-Bandwidth (GBW) and DC gain) and find the largest resistance you can tolerate for your settling constraints.
- Choose the switch MOSFET sizes.
- Pay attention to the switch resistance and the charge injection.

B. OTA Parameters

- Determine the OTA specifications using the ideal OTA model provided to you under CourseWorks/Files & Resources (gain, bandwidth, phase margin, slewing) based on the settling requirements.
- Use the OTA model and MOSFET switches derived in Section A and verify the functionality of your design.
- Make sure that you get a settling error of less than $V_{LSB}/2$ when you input a full-scale amplitude step.

2. OTA Transistor-Level Design (Week of May 4th, 2026)

- Design an OTA that satisfies your requirements: choose the topology, size transistors, verify specs.
- Make sure the input referred noise is of the same order of magnitude as the noise of the switch.
- Pay a lot of attention to the design of the common mode feedback circuit for your OTA that has sufficient bandwidth as well as phase margin.

2. Complete Design (May 13th, 2026)

- At this point, if you are unable to reach the specs as decided by phase I of your project, assume a reasonable set of parameters for the rest of the project. If you do change the specs, consult the TA for a sanity check. If you need any help with the OTA design, do not hesitate to approach the TA.
- Plug in the OTA in the fully differential switched capacitor sample-and-hold amplifier with MOSFET switches and run through the clock phases. Prove the 2x amplification happens with the right accuracy. Show that you have enough settling time, etc.
- Make sure that you get a settling error of $<V_{LSB}/2$ when you input a full-scale amplitude step.
- You need to show results for the following four cases at least:
 1. Apply a small signal, show desired amplification occurs.
 2. Apply at the input a small signal step going from V_{LSB} to $-V_{LSB}$ and back to V_{LSB} .
 3. Apply a slow ramp, covering the full-scale range of the design $[-V_{FS}, V_{FS}]$.
 4. Apply at the input a large signal step going from V_{FS} to $-V_{FS}$ and back to V_{FS} .

Remarks

1. As an input source you can use a voltage source and as a load assume a similar stage as your own stage. Keep in mind the requirement this imposes: the input and output common mode voltages need to be equal!
2. You need to design the OTA, capacitors and switches only. You can use the analogLib source vpulse to generate all your clocks; otherwise, for a simple non-overlapping clock generator refer to Fig. 14.3 in Johns and Martin (Second Edition).

Acknowledgment

This document has been developed by several TAs under the supervision of Columbia Professors through projects assigned to different offerings of this course.